AI-Integrated Assessment Integrity Lifecycle

Embedding Responsible AI Governance Throughout the Assessment Cycle

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About this Framework

The **AI-Integrated Assessment Integrity Lifecycle** provides a practical process framework for embedding responsible AI governance throughout the assessment lifecycle. Rather than treating academic integrity as an issue addressed only after assessment submission, the lifecycle positions integrity as a continuous educational responsibility that begins with assessment planning and extends through design, implementation, evaluation, moderation, and continuous enhancement.

Building upon the conceptual principles established in *From Assistance to Authorship*, the lifecycle integrates responsible AI governance into routine assessment practice. It encourages institutions to move beyond reactive approaches centred on AI detection or misconduct procedures by embedding transparency, authentic assessment design, professional judgement, and continuous improvement within everyday educational processes.

The framework is intended for lecturers, programme leaders, assessment designers, quality assurance teams, academic integrity professionals, educational developers, and institutional leaders responsible for designing, delivering, reviewing, and enhancing assessment. It also serves as the organisational framework that connects the companion resources into a coherent institutional approach to AI-assisted assessment.

Executive Summary

Artificial intelligence is reshaping every stage of assessment in higher education. Most institutional responses have concentrated on isolated interventions such as AI detection, disclosure statements, or post-submission investigations. While these measures remain important, they address only individual points within a much broader assessment system.

This document proposes an alternative perspective.

Assessment integrity should not be managed at the point of submission alone. Instead, responsible AI governance should be embedded throughout the entire assessment lifecycle—from initial assessment planning to institutional quality enhancement.

The AI-Integrated Assessment Integrity Lifecycle provides a structured governance model organised around nine interconnected stages:

1. Assessment Planning
2. Assessment Design
3. Student Preparation
4. AI Guidance and Disclosure
5. Assessment Completion

6. Submission and Verification
7. Marking and Feedback
8. Moderation
9. Review and Continuous Enhancement

Rather than introducing additional administrative procedures, the lifecycle integrates responsible AI governance into existing assessment practice. At each stage, educators are encouraged to consider how assessment design, communication, evidence of learning, and professional judgement can preserve academic integrity while recognising AI as an increasingly normal component of higher education.

The lifecycle complements the conceptual framework presented in *From Assistance to Authorship*. Whereas the conceptual framework explains how AI-assisted academic work should be evaluated, the present document explains how institutions can operationalise those principles throughout the assessment process.

Unlike institutional guidance that concentrates primarily on AI detection or academic misconduct procedures, the AI-Integrated Assessment Integrity Lifecycle positions responsible AI governance as a continuous educational process embedded throughout assessment design, delivery, evaluation, and enhancement. It therefore provides institutions with a practical implementation model for operationalising the principles established in *From Assistance to Authorship*.

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1. Introduction

Generative AI has fundamentally altered the environment within which assessment operates. Traditional approaches to academic integrity were largely designed for assessments completed independently by students using limited external assistance. Contemporary assessment takes place within a learning environment where sophisticated AI systems are readily available throughout planning, drafting, revision, and submission.

Many institutional responses continue to focus primarily on the submission stage. Students submit work, declarations are reviewed, AI detectors may be consulted, and concerns are investigated where appropriate.

Although necessary, this approach is incomplete.

Assessment integrity is not created at submission.

It begins long before students start writing and continues long after marks have been awarded.

Assessment planning influences opportunities for inappropriate AI use.

Assessment design determines which intellectual activities are evaluated.

Student guidance shapes expectations.

Verification confirms understanding.

Moderation supports consistency.

Quality review informs future assessment improvement.

Every stage contributes to assessment integrity.

The lifecycle presented here recognises this broader perspective. Rather than treating AI as a problem requiring isolated interventions, it integrates responsible AI governance across the complete assessment process.

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2. Why a Lifecycle?

Universities already manage assessment through interconnected systems rather than isolated events.

Assessment normally progresses through a sequence of activities.

An assessment is planned.

Tasks are designed.

Students receive guidance.

Learning takes place.

Work is submitted.

Assessment is marked.

Results are moderated.

Feedback informs future redesign.

Academic integrity should follow the same logic.

If AI governance occurs only after submission, institutions respond to problems rather than designing assessments that minimise them.

The lifecycle therefore adopts a preventative rather than reactive philosophy.

Instead of asking

How do we detect inappropriate AI use?

it asks

How do we design assessment systems in which responsible AI use becomes the educational norm?

This shift reframes assessment integrity as a continuous educational responsibility rather than a reactive compliance process. Governance becomes embedded throughout assessment rather than concentrated at the point of submission.

3. Guiding Principles

The lifecycle is governed by five overarching principles.

Learning before policing

Assessment should primarily support learning.

Academic integrity procedures protect educational standards but should not become the dominant purpose of assessment.

Prevention before investigation

Well-designed assessments reduce opportunities for inappropriate AI use.

Educational design is generally more effective than post-submission investigation.

Professional judgement

No automated system can determine authorship reliably.

Professional academic judgement remains central throughout the assessment process.

Transparency

Students should understand institutional expectations before beginning assessment.

Clear guidance reduces uncertainty and promotes responsible AI use.

Continuous improvement

Assessment governance should evolve alongside educational practice and technological development.

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4. The Assessment Integrity Lifecycle

The lifecycle consists of nine connected stages.

Each stage contributes differently to responsible AI governance.

Together they create a coherent institutional approach rather than isolated interventions.

Stage 1 — Assessment Planning

Assessment integrity begins before an assessment exists.

Programme teams should identify:

- intended learning outcomes;
- graduate capabilities;
- disciplinary expectations;
- appropriate role of AI;
- verification requirements.

Key question:

What intellectual capability is this assessment intended to evaluate?

Only after answering this question can appropriate AI guidance be developed.

Stage 2 — Assessment Design

Assessment should make intellectual engagement visible.

Design principles include:

- authentic tasks;
- contextual application;
- critical judgement;
- staged assessment;
- reflection;
- oral components where appropriate.

Assessment should reward thinking rather than text production.

Stage 3 — Student Preparation

Students require explicit guidance.

This includes:

- acceptable AI use;
- unacceptable AI use;
- disclosure expectations;
- examples;
- institutional policy;
- academic support.

Students should understand expectations before beginning work rather than discovering them during misconduct investigations.

Stage 4 — AI Guidance and Disclosure

Disclosure should be integrated into assessment rather than added retrospectively.

Students should know:

- whether disclosure is required;

- what should be disclosed;
- how disclosure influences marking;
- how AI use will be evaluated.

Disclosure supports transparency rather than punishment.

Stage 5 — Assessment Completion

Students complete their work while exercising intellectual responsibility.

The assessment should encourage:

- independent judgement;
- critical evaluation;
- reflective decision-making;
- appropriate AI use;
- documentation of significant AI assistance where required.

Stage 6 — Submission and Verification

Submission marks the transition from learning to formal academic evaluation. Traditionally, this stage has been the primary focus of academic integrity processes. Students submit work, declarations are completed, similarity reports may be reviewed, and concerns are investigated where appropriate.

Within an AI-enabled educational environment, however, submission should not be regarded as the point at which academic integrity begins. Rather, it provides an opportunity to confirm that the evidence collected throughout the assessment process supports confidence in the authenticity of the submitted work.

Verification should therefore be proportionate, educational, and evidence-based.

Routine assessment should proceed without unnecessary administrative burden. Additional verification should be considered only where legitimate academic reasons exist, such as unexplained inconsistencies between drafts and the final submission, substantial changes in writing quality, contradictory evidence emerging during supervision, or assessment tasks where confirmation of intellectual ownership forms an expected component of the learning outcomes.

Verification methods may include:

- review of assessment drafts;
- reflective statements;
- annotated research notes;
- process portfolios;
- revision histories;
- oral verification where appropriate;
- discussion with the student.

Importantly, these activities should not be interpreted as investigations. Their primary purpose is to strengthen confidence that the submitted work accurately represents the student's intellectual contribution.

Key Question

Does the available evidence support confidence that the submitted work represents the student's own intellectual engagement?

Stage 7 — Marking and Feedback

Marking should evaluate achievement of learning outcomes rather than merely the characteristics of the submitted text.

Generative AI makes this distinction increasingly important.

Markers should avoid assuming either that polished writing demonstrates independent scholarship or that fluent writing necessarily indicates inappropriate AI use. Instead, evaluation should remain focused on evidence of disciplinary understanding, analytical reasoning, methodological justification, and intellectual judgement.

Where AI has been disclosed appropriately, the disclosure itself should not normally influence academic grades. Students should be evaluated according to the quality of their learning, not according to whether AI was used responsibly within institutional expectations.

Feedback should continue to promote learning by recognising strengths while identifying opportunities for further intellectual development. Where AI use appears to have limited deeper engagement with the assessment task, feedback should address the educational implications rather than focusing exclusively upon policy compliance.

Examples include encouraging students to:

- explain reasoning more explicitly;
- strengthen critical evaluation;
- justify methodological decisions;
- demonstrate greater independence of thought;
- distinguish personal interpretation from generated summaries.

Key Question

Does the submitted work demonstrate the learning outcomes the assessment was designed to evaluate?

Stage 8 — Moderation

Moderation plays a central role in maintaining fairness and consistency across programmes. AI introduces new forms of variation that moderation processes must address.

Different markers may hold different assumptions regarding acceptable AI use. Some may interpret extensive language support as educationally appropriate, while others may regard similar practices as problematic. Without shared principles, comparable work may receive inconsistent academic judgements.

The lifecycle therefore encourages moderation discussions that focus upon common evaluative principles rather than individual AI tools.

Moderation meetings should consider questions such as:

- Were expectations communicated consistently?
- Were disclosure requirements applied fairly?
- Did markers distinguish between supportive and substitutive AI functions?
- Were concerns regarding authorship supported by evidence rather than intuition?
- Were professional judgements applied consistently across the cohort?

Moderation should also identify recurring patterns that may inform future assessment design, staff development, or institutional guidance.

Key Question

Were evaluative decisions consistent with institutional expectations and applied fairly across students?

Stage 9 — Review and Continuous Enhancement

Assessment integrity does not conclude when grades are released.

Each assessment provides valuable evidence that can inform future practice.

Programme teams should periodically review:

- student performance;
- effectiveness of assessment design;
- recurring AI-related issues;
- staff experiences;
- moderation outcomes;
- student feedback;
- developments in AI technologies;
- external regulatory expectations.

Rather than responding to each technological development through new regulations, institutions should gradually refine assessment practice while preserving enduring educational principles.

Continuous enhancement also recognises that AI governance remains an evolving field. Universities should expect to revise guidance as professional understanding develops, ensuring that policy continues to support learning rather than merely responding to technological change.

Key Question

What has this assessment taught us about improving future assessment practice?

5. Roles and Responsibilities

Successful implementation requires shared responsibility across the institution. AI governance should not be regarded solely as the responsibility of academic integrity offices or individual lecturers.

Lecturers

Lecturers are responsible for designing assessments that make intellectual engagement visible, communicating expectations clearly, applying professional judgement consistently, and supporting responsible AI literacy.

Programme Leaders

Programme leaders ensure consistency across modules by coordinating assessment approaches, reviewing programme-level coherence, and supporting moderation processes.

Assessment Designers

Assessment designers contribute by aligning assessment methods with intended learning outcomes and considering how AI may influence evidence of student achievement.

Quality Assurance Teams

Quality assurance teams monitor consistency, review institutional practices, support enhancement initiatives, and ensure that governance remains aligned with educational objectives.

Academic Integrity Staff

Academic integrity specialists provide guidance on policy interpretation, support investigations where necessary, contribute to staff development, and promote institution-wide consistency.

Students

Students remain responsible for maintaining intellectual ownership of their work, using AI responsibly, complying with disclosure expectations, and engaging honestly with assessment processes.

Assessment integrity is therefore understood as a shared institutional responsibility rather than a compliance obligation imposed upon individual students.

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6. Integrating the Companion Resources

The lifecycle is designed to operate alongside the companion framework suite.

Different resources become relevant at different stages of the assessment process.

Assessment Lifecycle Stage	Companion Resource
Assessment Planning	From Assistance to Authorship
Assessment Design	Assessment Authenticity Checklist
Student Preparation	AI Disclosure Framework
Assessment Completion	AI Risk-Level Matrix
Submission & Verification	Oral Verification Toolkit
Moderation & Review	Institutional AI Governance Guide

Together these documents create an integrated governance ecosystem rather than a collection of independent resources.

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7. Implementation Roadmap

Institutions adopting the lifecycle should implement it gradually rather than attempting comprehensive policy reform simultaneously.

A staged approach is recommended.

Phase 1

Develop institutional understanding.

- staff development;
- review existing policies;
- identify assessment priorities.

Phase 2

Embed lifecycle thinking.

- redesign selected assessments;
- introduce disclosure guidance;
- strengthen moderation.

Phase 3

Integrate institution-wide.

- programme-level consistency;
- quality assurance review;
- ongoing evaluation.

Implementation should remain proportionate to institutional context while preserving the core principles established throughout the lifecycle.

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8. Institutional Implementation Checklist

The following checklist is intended to support programme leaders, assessment committees, and quality assurance teams when embedding the AI-Integrated Assessment Integrity Lifecycle within institutional practice.

Completion of every item is not expected immediately. Instead, the checklist provides a structured framework for continuous enhancement.

Area	Evidence
Assessment learning outcomes clearly identify the intellectual capabilities being assessed.	☐
Programme teams have discussed the appropriate role of AI within assessments.	☐
Students receive explicit guidance on acceptable AI use before beginning assessment.	☐
Disclosure requirements are proportionate and clearly communicated.	☐
Assessment tasks encourage authentic intellectual engagement rather than text production alone.	☐
Verification approaches are educational, proportionate, and evidence-based.	☐
Marking criteria focus upon evidence of learning rather than assumptions regarding AI use.	☐
Moderation processes include discussion of AI-related assessment decisions.	☐
Assessment outcomes are reviewed to inform future redesign.	☐

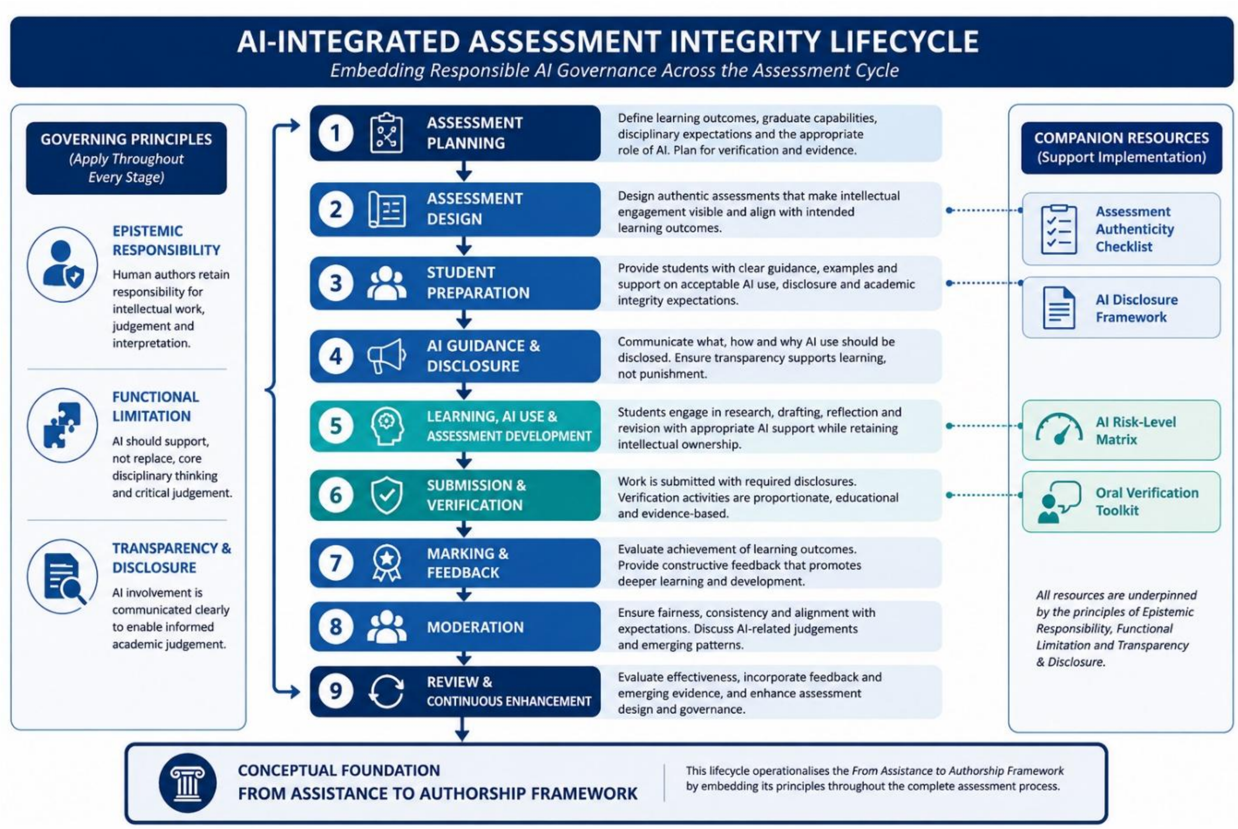
Area	Evidence
Staff receive continuing professional development regarding responsible AI assessment practices.	☐

Institutions are encouraged to adapt this checklist to local policies while preserving the principles of responsibility, functional limitation, and transparency established within the companion conceptual framework.

Appendix A

The AI-Integrated Assessment Integrity Lifecycle

Figure 1



Caption

The AI-Integrated Assessment Integrity Lifecycle illustrates how responsible AI governance should be embedded throughout the complete assessment process. Rather than concentrating academic integrity at the point of submission, the lifecycle integrates governance across planning, design, student preparation, assessment completion, verification, marking, moderation, and continuous enhancement. Each stage contributes differently to maintaining confidence in authentic student learning while recognising AI as a normal component of contemporary higher education.

Appendix B

Summary of Assessment Lifecycle Stage

Stage	Primary Purpose	Key Question
Assessment Planning	Define educational purpose	What capability is being assessed?
Assessment Design	Create authentic assessment	Does the assessment reward thinking?
Student Preparation	Communicate expectations	Do students understand responsible AI use?
AI Guidance & Disclosure	Promote transparency	What should students disclose?
Assessment Completion	Support responsible learning	Has the student retained intellectual ownership?
Submission & Verification	Confirm authenticity	Does available evidence support authorship?
Marking & Feedback	Evaluate learning	Have learning outcomes been demonstrated?
Moderation	Ensure consistency	Were decisions fair and consistent?
Review & Enhancement	Improve future assessment	What should change next time?

Appendix C

Relationship with the Companion Framework Suite

This lifecycle should be read alongside the accompanying framework documents.

Companion Document	Primary Function
From Assistance to Authorship	Conceptual foundation for evaluating AI-assisted academic work

Companion Document	Primary Function
AI Disclosure Framework	Supports proportionate transparency regarding AI assistance
Oral Verification Toolkit	Provides structured approaches to confirming intellectual ownership
Assessment Authenticity Checklist	Supports assessment design promoting visible intellectual engagement
AI Risk-Level Matrix	Assists evaluation of supportive and substitutive AI functions
Institutional AI Governance Guide for Assessment Integrity	Supports institution-wide implementation and quality assurance

Together, these resources provide a coherent governance ecosystem. Each document addresses a different aspect of AI-integrated assessment while maintaining a common conceptual language.

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9. Limitations

The AI-Integrated Assessment Integrity Lifecycle is intended as a conceptual and practice-informed implementation model rather than a mandatory institutional procedure.

Several limitations should therefore be recognised.

First, the lifecycle does not prescribe identical assessment practices across all disciplines. Different fields of study require different forms of evidence, assessment design, and AI integration.

Second, implementation will necessarily vary according to institutional regulations, national quality assurance frameworks, professional accreditation requirements, and available resources.

Third, the lifecycle assumes continuing professional judgement by educators. It is not intended to replace academic expertise through algorithmic decision-making or automated governance.

Finally, while informed by published scholarship, professional practice, and ongoing framework development, the lifecycle has **not yet undergone formal empirical**

evaluation as an institutional governance model. Institutions adopting the framework should therefore regard implementation as part of an ongoing process of educational enhancement and organisational learning rather than as the application of a validated assessment instrument.

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10. Conclusion

Artificial intelligence has fundamentally changed the context in which assessment operates. Effective governance can no longer be achieved through isolated interventions applied only after work has been submitted.

Assessment integrity begins long before submission and continues long after marks have been awarded.

The AI-Integrated Assessment Integrity Lifecycle responds to this reality by embedding responsible AI governance throughout the complete assessment process. Rather than positioning AI as an external threat requiring continuous surveillance, the lifecycle recognises AI as an increasingly normal feature of academic practice while reaffirming the enduring importance of intellectual responsibility, authentic learning, and professional judgement.

The lifecycle therefore represents a shift from reactive enforcement towards proactive educational governance.

Its purpose is not to regulate technology itself.

Its purpose is to preserve confidence that assessment continues to evaluate meaningful human learning within an AI-enabled educational environment.

Framework Status

Framework type: Institutional implementation framework

Development status: Conceptual and practice-informed

Foundation: Developed from published scholarship, professional practice in higher education, and the companion conceptual framework *From Assistance to Authorship*.

Empirical status: This lifecycle has not yet undergone formal empirical validation as an institutional governance model. Future evaluation is recommended before claims regarding institutional effectiveness are made.

Version History

Version	Date	Description
1.0	July 2026	First public release